Life History Strategies in an Explicit Patch Age Model

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Introduction

A mechanistic understanding of species coexistence remains a significant challenge in ecological theory. Classical forest ecology hypotheses suggest that disturbances and patch dynamics enable coexistence among species with various life history strategies, termed "successional niche" differentiation [2]. However, prior mathematical models have not fully delineated how variations in life history strategies can promote coexistence under disturbance and succession. We extend a PDE model that includes explicit patch aging, disturbances, and within-patch competition, by incorporating the age of reproductive maturity as a key trait. We find several trade-offs involving traits associated with reproduction and mortality. We also find species abundance patterns that differentiate these mechanisms from classical succession. We use ForestGEO plot data from BCI to look for empirical evidence of these trade-offs.

Model

Patch Dynamics

To model patch aging with disturbance we use the McKendrick - Von Foerster Equation,

$$\begin{cases} \frac{\partial \rho}{\partial t} + \frac{\partial \rho}{\partial a} = -\gamma(a)\rho(a,t) \\ \rho(a,0) = \rho_0(a) \\ \rho(0,t) = \int_0^\infty \gamma(a)\rho(a,t)\, \mathrm{d}a \end{cases}$$

where $\rho(a,t)$ represents the proportion of patches of age a at time t, and γ is the rate of disturbance. Disturbance corresponds to the "death" of a patch. The boundary condition, $\rho(0,t)$, represents patch ages being reset to $a=0$ after a disturbance.

Competitive Dynamics

We let $n_i(a,t)$ represent the number of individuals of species i on patches of age a at time t. Let τ_i be the age of first reproduction. The generic within patch competitive dynamics can be written as,

$$\begin{cases} \frac{\partial n_i}{\partial t} = \underbrace{-\frac{\partial n_i}{\partial a}}_{aging} + H(a-\tau_i)[\overbrace{r_i(a,t-\tau_i;\rho(a),\vec{n}(a,t))}^{reproduction} - \underbrace{\mu_i(a,t;\vec{n}(a,t))}_{mortality}] \\ n_i(a,0) = n_{i,o}(a) \\ n_i(a,t) = 0 \quad \text{for } a \le \tau_i \end{cases}$$

The density dependence can act on either component. We focus on the following cases:

1. Competition on Reproduction:

$$r = b_i \int_{\tau_i}^{\infty} \rho(a', t-\tau_i) n_i(a', t-\tau_i)\left(1 - \alpha_i \sum_k n_k(a', t-\tau_i)\right) da'$$

$$\mu = \mu_i n_i(a,t)$$

2. Competition on Mortality.

$$r = b_i \int_{\tau_i}^{\infty} \rho(a', t-\tau_i) n_i(a', t-\tau_i)\, da'$$

$$\mu = \mu_i n_i(a,t)\left(1 + \alpha_i \sum_k n_k(a,t)\right)$$

The patch age distribution is coupled to n_i in the reproductive process since *all* individuals older than τ can reproduce. We choose competition coefficients, α_k, so that if you remove disturbance and patch structure you cannot get coexistence.

Methods

To study coexistence, we derive criteria for *feasible coexistence*, *mutual invasion*, and *local stability*. *Feasibility* ensures positive reproduction at equilibrium:

$$r_i(n_i^*, n_j^*) > 0, \quad r_j(n_i^*, n_j^*) > 0$$

Mutual invasion ensures a species can increase from rarity while another species is at equilibrium. We derive an ODE for total density, $\langle n_i \rangle = \int \rho(a) n_i(a')\, da'$, and determine when the per-capita growth rate is positive:

$$\frac{1}{\langle n_i \rangle}\frac{\mathrm{d}\langle n_i \rangle}{\mathrm{d}t}\bigg|_{\epsilon(a), n_j^*(a)} > 0$$

For *local stability*, we use an asymptotic argument to find when perturbations decay and the system returns to equilibrium.

We examine when trade-offs in the time of first reproduction, τ, enable coexistence. For example, if species i has a later first reproduction age ($\tau_i = \tau_j + \Delta_\tau$ with $\Delta\tau > 0$) and birth rate ($b_i = b_j + \Delta_b$), we then take feasibility criteria and derive equivalent conditions to show a trade-off is necessary for coexistence:

$$r_i(\tau_j + \Delta_\tau, b_j + \Delta_b; n_i^*, n_j^*) > 0, \quad r_j(\tau_j, b_j; n_i^*, n_j^*) > 0$$

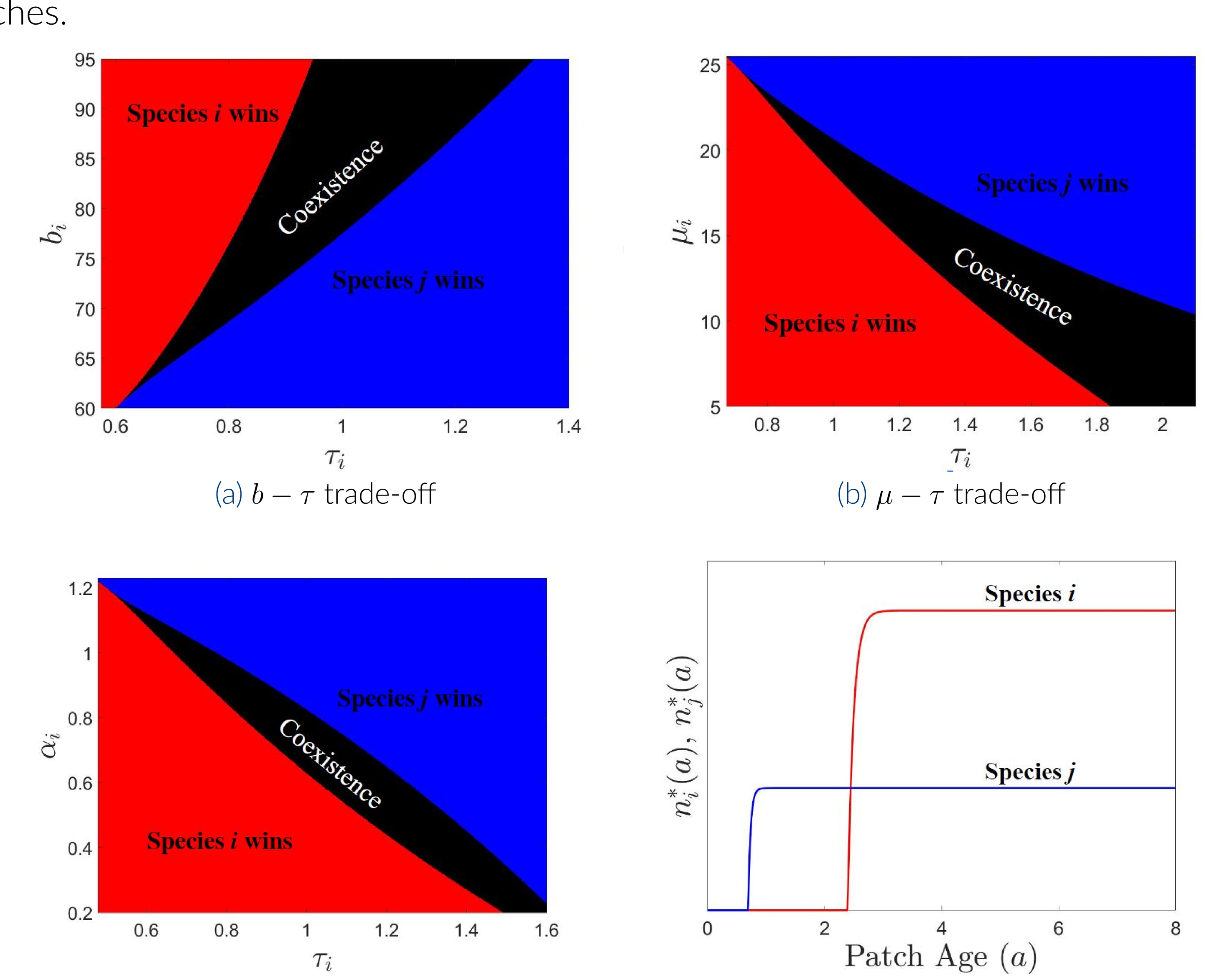
$$\iff \underbrace{0 < f_1(\vec{b}, \vec{\mu}, \vec{\alpha}, \gamma, \Delta_\tau) < \Delta_b}_{\text{Trade-off is Necessary}} \overbrace{\Delta_b < f_2(\vec{b}, \vec{\mu}, \vec{\alpha}, \gamma, \Delta_\tau)}^{\text{Region of Coexistence}}$$

Density Dependence on Reproduction

In this case, we have a general form for feasible coexistence,

$$\frac{\alpha_i \langle n_i^* n_j^* \rangle}{\alpha_j \langle n_j^{*2} \rangle} > \frac{\langle n_i^* \rangle - 1}{\langle n_j^* \rangle - 1} > \frac{\alpha_i \langle n_i^{*2} \rangle}{\alpha_j \langle n_i^* n_j^* \rangle}$$

We can determine n_k^* and compute these quantities allowing us to show equivalence to mutual invasion and to identify regions of coexistence. Figure 1d shows the species abundance pattern across patch ages when they coexist. We don't observe the typical "successional niche" structure, despite species i having exclusive access to younger patches.

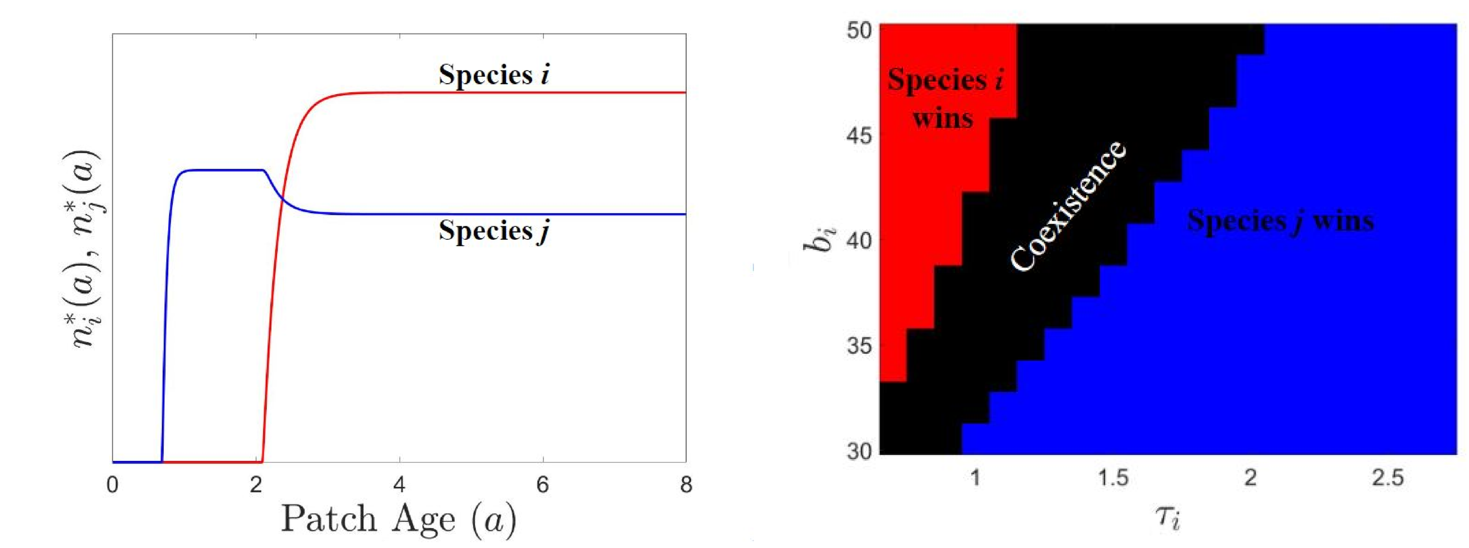


(a) $b-\tau$ trade-off (b) $\mu-\tau$ trade-off (c) $\alpha-\tau$ trade-off (d) Abundance Patterns in $b-\tau$ trade-off

Figure 1. a)-c) Regions of Coexistence generated by Life History Strategies, d) General Solution to PDE.

Density Dependence on Mortality

In this case, we cannot analytically determine $n_k^*(a)$ or write explicit conditions for feasible coexistence and mutual invasion. Instead, we numerically solve the PDE across various parameters to generate coexistence regions (Fig. 2b shows one of these trade-offs). Fig. 2a shows a partial successional structure where species j can initially exploit younger patches but is not entirely outcompeted, unlike typical successional dynamics.



(a) Coexistence with partial successional niche (b) $b-\tau$ trade-off

Figure 2. a) General Solution to PDE, b) $b-\tau$ coexistence region.

Empirical Evaluation

Using BCI ForestGEO Plot data [1], we estimate the time to reproductive maturity and sapling mortality at the species level. Figure 3 shows a negative relationship between these traits, indicating that as age of reproductive maturity increases, sapling mortality decreases. A boundary line suggests a b-τ trade-off with limits on trait combinations. The age of first reproduction is estimated as $\tau = \ln(\frac{1}{2}\mathrm{DBH}_{\mathrm{max}})/\mathrm{RGR}$, with RGR calculated individually from 1 cm to $\frac{1}{2}\mathrm{DBH}_{\mathrm{max}}$ and averaged across species.

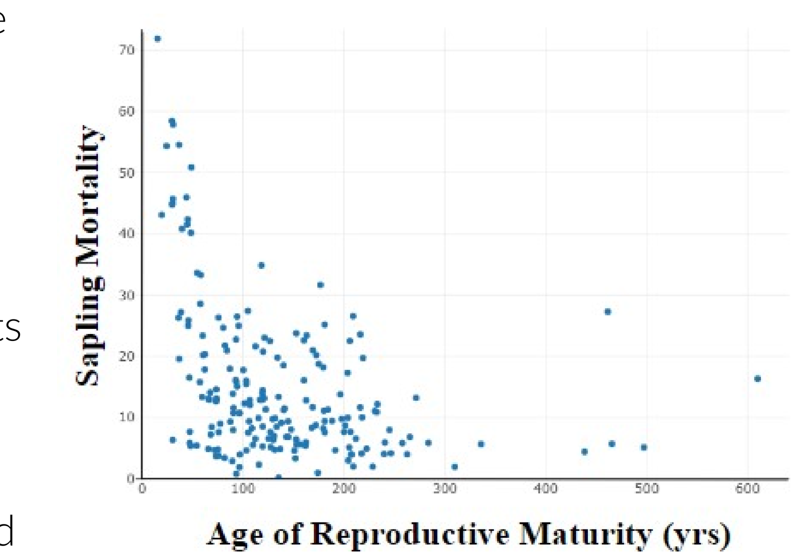


Figure 3

Conclusion

We find several trade-offs between age of first reproduction and demographic traits related to reproduction and mortality that generate coexistence. Figures 1a and 2a highlight the absence of the typical successional niche structure. Our model provides a framework for understanding landscape-level coexistence from explicit life history variations on a within patch level with disturbance being a necessary ingredient.

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Acknowledgement: The BCI forest dynamics research project was made possible by National Science Foundation grants to Stephen P. Hubbell, support from ForestGEO, STRI, the John D. and Catherine T. MacArthur Foundation, the Mellon Foundation, the Small World Institute Fund, and numerous private individuals, and through the hard work of over 100 people from 10 countries over the past three decades.

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